TOPOLOGICAL ORIGIN OF THE FERMION MASS HIERARCHY AND THE KOIDE FORMULA ON FRACTAL BOUNDARIES

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Abstract. The empirical Koide formula accurately relates the charged lepton masses but lacks a foundational derivation. In this paper, we resolve this using the Unified Grand Synthesis framework. By modeling spacetime as a fractional Sierpinski tensor network, fermion flavor space corresponds to the $3D$ representation of the boundary's permutation symmetry. To preserve $SU(2)_L$ chiral gauge invariance, this symmetry forces the emergent *Yukawa coupling matrix* to be strictly circulant. We demonstrate that in the 2D Holographic UV limit, particle exchanges are governed by the Braid Group B_3, inducing a complex topological phase. As the network percolates to macroscopic 3D space, the Braid Group collapses to S_3 (saving the Spin-Statistics Theorem), but the complex phase is topologically frozen into the Hermitian Yukawa matrix, providing a rigorous geometric origin for CP Violation. Upon electroweak symmetry breaking, this yields the Koide ratio $K = 2/3$ as an exact IR fixed point for color-singlet charged leptons, while Color-Flavor Entanglement explicitly modifies it for quarks.

1. Introduction

The Koide formula for charged leptons ($K \approx 2/3$) is remarkably precise at low energies [2]. If the mass hierarchy is driven by spacetime topology, it must reconcile with the chiral gauge symmetries of the electroweak model [3], the Spin-Statistics Theorem, and account for the CP violation necessary for baryogenesis [4]. In this work, we prove that the exact Koide ratio and CP violation are geometric signatures of a Complex Circulant Yukawa Matrix on the fractal vacuum.

2. Flavor Symmetry, UV Braiding, and CP Violation

We model spacetime as a Sierpinski Tensor Network [1]. The macroscopic boundary possesses 3 vertices, generating an S_3 permutation symmetry. Elementary fermions exist on this boundary; their flavor space $V_{\text{flavor}} \cong \mathbf{1} \oplus \mathbf{2}$ guarantees exactly three generations.

To preserve electroweak chiral symmetry, the permutation topologically constrains the interactions between chiral fermions and the emergent scalar Higgs field, generating a circulant Yukawa matrix Y_{circ}.

Theorem 2.1 (Holographic Phase Freezing and CP Violation)**.** *At the deep UV Planck scale, the fractal boundary acts as a strict 2-dimensional topological space. Here, fermion exchange paths entangle, elevating the symmetry to the* ***Braid Group*** B_3*. Exchanges acquire a non-trivial anyonic phase* $e^{i\delta}$*. The resulting Yukawa coupling matrix is* ***Complex Hermitian****. As the geometry percolates to the macroscopic IR limit (3D space), the Braid Group mathematically collapses to the Permutation Group* S_3*, restoring standard fermion Spin-Statistics (Pauli Exclusion). However, the complex phase* δ *is topologically "frozen" into the algebraic matrix elements of* Y_{circ}*. This fossilized fractional phase provides the rigorous geometric origin for* ***CP Violation*** [4].

3. Electroweak Symmetry Breaking and Universality

When the geometry percolates, the Higgs field acquires a Vacuum Expectation Value v. The physical mass matrix is dynamically generated: $M = Y_{circ}\frac{v}{\sqrt{2}}$. Because the VEV is a real scalar, the algebraic topology of the complex circulant matrix is inherited by the physical masses. The Fractional Dirac Operator eigenvalues $\lambda_i \propto \sqrt{m_i}$ algebraically dictate the invariant $K = 2/3$.

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Lemma 3.1 (Gauge-Flavor Splitting)**.** *The unperturbed circulant Yukawa symmetry holds strictly for particles decoupled from topological background fields.*

- ***Charged Leptons ($SU(3)_C$ Singlets):*** *Neutral under the strong force, leptons preserve the pure representation. Their mass matrix flows to the $K = 2/3$ IR fixed point.*
- ***Quarks ($SU(3)_C$ Triplets):*** *Gluon exchanges on the fractal boundary induce* ***Color-Flavor Entanglement****, breaking the circulant symmetry and heavily modifying K.*

4. Conclusion

The three generations, the Koide hierarchy, and CP violation are fundamental consequences of the Sierpinski boundary topology. The UV Braid Group B_3 induces the complex phase for matter-antimatter asymmetry, which is frozen during IR percolation to preserve the Spin-Statistics Theorem. We proved that the Koide 2/3 ratio is a topologically protected IR fixed point exclusively for charged leptons.

AI Ethics and Tool Use Statement

Generative AI tools (large language model assistants) were employed solely for grammar checking, adversarial mathematical cross-verification, and LaTeX formatting assistance during drafting. The conceptual framework, geometric theorems, mathematical formulations, and conclusions are entirely the original intellectual work of the author, who assumes full academic responsibility for the integrity and accuracy of the contents.

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